Adaptive Regularized Low-Rank Tensor Decomposition for Hyperspectral Image Denoising and Destriping

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***Abstract*— Hyperspectral images (HSIs) are inevitably degraded by a mixture of various types of noise, such as Gaussian noise, impulse noise, stripe noise, and dead pixels, which greatly limits the subsequent applications. Although various denoising methods have already been developed, accurately recovering the spatial−spectral structure of HSIs remains a challenging problem to be addressed. Furthermore, serious stripe noise, which is common in real HSIs, is still not fully separated from the previous models. In this article, we propose an adaptive hyper-Laplacian regularized low-rank tensor decomposition (LRTDAHL) method for HSI denoising and destriping. On the one hand, the stripe noise is separately modeled by the tensor decomposition, which can effectively encode the spatial–spectral correlation of the stripe noise. On the other hand, adaptive hyper-Laplacian spatial–spectral regularization (AHSSTV) is introduced to represent the distribution structure of different HSI gradient data by adaptively estimating the optimal hyper-Laplacian parameter, which can reduce the spatial information loss and over-smoothing caused by the previous total variation (TV) regularization. The proposed model is solved using the alternating direction method of multipliers (ADMM) algorithm. Extensive simulation and real-data experiments all demonstrate the effectiveness and superiority of the proposed method.**

***Index Terms*— Adaptive hyper-Laplacian, denoising and destriping, hyperspectral images (HSIs), tensor completion.**

I. Introduction

HYPERSPECTRAL images (HSIs) obtained by imaging spectroscopy contain hundreds of spectral bands, providing rich spectral information. Taking advantage of the abundant spatial and spectral information, HSIs have been widely applied in various applications, such as mineral exploration, precision farming, environmental protection [1], [2], and so on. However, the narrower spectral channels usually mean that the sensors receive fewer signals, making HSIs highly susceptible to interference from various types of noise, such as stripe noise, impulse noise, Gaussian noise, deadlines, and dead pixels [3], [4]. Consequently, developing effective HSI denoising methods is a challenging and necessary task for the various subsequent applications.

In recent decades, considerable progress has been made in the field of HSI denoising. The straightforward approach is to treat each band of the HSI as a 2-D grayscale image and then adopt a traditional grayscale image denoising method [4], such as block-matching and 3-D filtering (BM3D) [5] or weighted nuclear norm minimization (WNNM) [6] However, these approaches obtain relatively poor restoration performances since they ignore the strong spectral correlation of HSIs. Machine learning and variational-based methods are the commonly used methods that can use the spectral correlation of HSIs. Due to the formidable nonlinear fitting capability of deep learning, a number of studies have integrated deep learning into HSI denoising [7], [8], [9], [10], [11], [12], [13], [14], [15]. However, due to the complex interplay of the multiple factors contributing to the degradation of real-world HSIs, the current deep learning-based methods for HSI denoising can struggle to accurately replicate this intricate physical process [14]. Variational-based methods can efficiently integrate the various prior knowledge of the HSI and noise, allowing for a more flexible adaptation to different noise scenarios. Recently, HSI denoising methods based on the variational framework considering the spectral correlation of HSIs have emerged as mainstream techniques, mainly including low-rank matrix decomposition methods and low-rank tensor decomposition methods.

The low-rank matrix decomposition-based methods have shown remarkable performances. Typically, these methods first unfold the HSI along the spectral dimension into a 2-D matrix. A denoising technique is then applied to the 2-D matrix. The low-rank matrix decomposition-based methods are based on two facts, i.e., low-rank matrix decomposition can encode the high spectral correlation of HSIs well, and impulse noise, stripe noise, and deadlines all show a sparse characteristic. The low-rank matrix recovery (LRMR) method [16] lexicographically rearranges the 3-D HSI into 2-D Casorati matrices whose

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columns are made up of vectorized bands. It then approximates the low-rank matrix decomposition by minimizing the nuclear norm of the Casorati matrix. The LRMR method is adept at capturing the low-rank redundancy (LR) of HSI but lacks consideration for its local smoothness (LSS). The 3-D correlated total variation regularization (3DCTVRPCA) model [17] systematically explores and encodes the joint prior expressions of LR and LSS. Specifically, in the gradient domain of HSI, this model constructs Casorati matrices and performs matrix decomposition. Building upon this foundation, the global variation and local structure low-rank (GLLR) model [18] utilizes the local structure low-rank (LLR) characteristics of HSI to construct Casorati matrices. Leveraging the low-rank features of Casorati matrices, the model successfully captures and restores the local structural features of HSI.

However, the low-rank matrix decomposition methods exploit the low-rank characteristic from the perspective of the spectrum, which results in insufficient utilization of the spatial information and suboptimal denoising results in the presence of severe noise. To alleviate this problem, several effective spatial prior regularization terms have been introduced into low-rank matrix decomposition models. For example, the low-rank total variation (LRTV) method [19] introduces total variation (TV) regularization into the LRMR model so that the TV regularization can efficiently preserve the spatial piecewise smoothness; the spatial–spectral TV (SSTV) regularization term extends the TV constraint to the spectrum and has achieved good results when combined with low-rank matrix decomposition [21]; texture-preserved TV (TPTV) regularization introduces a weighting scheme to the gradient maps of HSI, which help preserve the texture structure [20], and the nonlocal self-similarity regularization (NSS) term, which exploits the similarity between nonlocal patches and the similarity within patches simultaneously by applying low-rank regularization for the clustering of similar patches, has also been widely combined with a low-rank matrix decomposition model, such as global and nonlocal low-rank factorizations (GLF) [22] and subspace-based guided nonlocal low-rank approximation (SGNLR) [23]. However, all these matrix decomposition-based methods need to sort the 3-D data lexicographically into 2-D data, which inevitably results in spatial structural detail information loss.

Since HSIs are 3-D tensors, tensor-based denoising methods have received increasing attention as they can effectively utilize the global spatial–spectral correlation of the HSI while efficiently restoring the spatial structure. Recently, several methods based on different tensor decomposition schemes have been proposed, such as Tucker tensor decomposition [24], rank-1 tensor decomposition [25], tensor-singular value decomposition (t-SVD) [26], and tensor ring decomposition [27]. Satisfactory denoising results can be achieved with these methods in the preservation of spatial and spectral structures and the decomposition efficiency. In addition, the tensor-based methods are often combined with other priors to achieve better results in removing complex noise. The low-rank tensor decomposition-total variation (LRTDTV) method [28] combines tensor decomposition and the TV technique by utilizing the low-rank characteristic of spatial–spectral data and the LSS, and has demonstrated a pleasing performance. The tensor-correlated TV [29] method builds a unique regularization term, which essentially encodes both the global low-rank and LSS priors of the HSI simultaneously by applying tensor decomposition in the gradient domain. The global tensor low-rankness and nonlocal SVD-aided group sparsity (GTL_NSGS) [30] model imposes group sparsity regularization priors on the difference domain of nonlocal similar tensors. This joint expression effectively captures the spatial self-similarity, spectral low-rankness, and group sparsity in the difference domain of HSI.

Although significant successes have been achieved with the recent HSI denoising methods, there are still some challenges that need to be addressed. Firstly, the current studies have mainly focused on exploiting the spatial–spectral correlation of the image components using various methods, but less attention has been paid to the stripe noise structure, which is often modeled as sparse noise. In fact, stripe noise exhibits a high correlation in the spatial and spectral dimensions [31], [32], [33], [34], [35], and it spans most of the image and is no longer sparse, meaning that the current low-rank based methods cannot completely separate stripe noise from the clean HSI. The double low-rank (DLR) matrix decomposition method [3] models the stripe noise separately and encodes the spatial correlation of stripe noise by extending the LRMR method with per-band low-rank matrix decomposition regularization, obtaining great results in denoising and destriping. However, the DLR method is a matrix decomposition-based method, which neglects the correlation among the stripe noise in different bands. Therefore, it is necessary to fully depict the spatial–spectral correlation of stripe noise when designing an HSI denoising model.

Secondly, most of the current HSI denoising methods based on LSS priors use the TV regularization-based ℓ_1-norm [36], [37], [38], [39], [40], which cannot faithfully depict the distributional structure of gradient data and usually results in the heavy-tailed phenomenon. To address this issue, the ℓ_p norm has been used to replace the ℓ_1 norm to encode the spatial–spectral LSS of HSIs in recent research, in methods such as hyper-Laplacian regularized nonlocal low-rank matrix recovery (HyNLRMR) [41] and hyper-Laplacian spectral-spatial total variation (HTV) [42]. The probability density function (pdf) of the ℓ_p norm is the hyper-Laplacian distribution, which can fit image gradient data well and avoid the heavy-tailed phenomenon. However, the shape of the hyper-Laplacian distribution is mostly controlled by parameter p, with noise remaining in high p value conditions and over-smoothing in small p values. Since the distribution of the gradient across distinct HSIs is different, the optimal p value for hyper-Laplacian PDFs should also be different. Thus, obtaining the optimal p value when applying the ℓ_p norm for HSI denoising is a key issue, and it would represent great progress if we could design a regularization term that can adaptively adjust parameter p based on the characteristics of each HSI.

In this article, to alleviate these problems, we propose an adaptive hyper-Laplacian regularized low-rank tensor decomposition (LRTDAHL) method, which can remove several types

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of noise, particularly stripe noise, and maintain the spatial structure well at the same time. The main contributions of this work are as follows.

1) We integrate tensor low-rank modeling of both the HSI and stripe noise into a unified model and introduce tensor Tucker decomposition to encode the high spatial and spectral correlation in stripe noise, accurately separating it from the observed HSI.
2) We introduce an adaptive hyper-Laplacian spatial–spectral regularization (AHSSTV) term to encode the structural sparsity in the spatial and spectral dimensions of the HSI. Unlike the previous hyper-Laplacian regularization terms, AHSSTV can adaptively estimate the hyper-Laplacian parameter of the clean gradient from the observed HSI, thereby recovering the LSS of the HSI more finely.

II. LRTDAHL Method for HSI Mixed Denoising

A. Low-Rank Tensor Decomposition of Stripes and HSI

As is well known, HSIs are located in a low-dimensional space, with high redundancy in the spectral dimension. Its redundancy can be effectively represented using a low-rank tensor framework. For stripe noise in HSIs, this also has a clear low-rank feature. To further study the characteristics of stripe noise, we used the stripe generation method proposed in [31] to simulate a set of stripe noise so that we could study the correlation of stripes from a quantitative perspective. Fig. 1 shows the simulated stripe noise, the unfolding matrices of the spatial and spectral dimensions, and the corresponding singular value curves. The decay of the singular value curves of the unfolding matrices proves the strong correlation of stripe noise in the two spatial dimensions and the relatively weaker correlation in the spectral dimension. Based on the prior knowledge and mathematical research into the spatial–spectral correlation of tensor data, we introduce low-rank Tucker decomposition to aggregate the decomposed matrices obtained as described above

$$\mathcal{B} = \mathcal{U} \times_1 G_1 \times_2 G_2 \times_3 G_3 \tag{1}$$

where $\mathcal{U}$ is the abundance tensor and G_i is the endmember matrix. Compared to modeling stripe noise with per-band low-rank matrix decomposition, Tucker decomposition can effectively encode the low-rank information in both spatial dimensions, resulting in a more accurate extraction of stripe noise from HSIs that contain noise [35].

B. HSI Observation Model

The traditional processing methods typically decompose the obtained HSI with mixed noise $\mathcal{Y}$ into clean image $\mathcal{X}$, Gaussian noise $\mathcal{N}$, and sparse noise $\mathcal{S}$, with the stripe noise categorized as sparse noise $\mathcal{S}$. However, previous studies [31] have indicated that the generation and actual distribution of stripe noise are different from those of sparse noise, and they separated the stripe noise from $\mathcal{X}$ and modeled it separately as the matrix $\mathcal{B}$. Based on the analysis results presented in Section II-A, it is evident that stripe noise exhibits a certain level of correlation in both the spatial and spectral dimensions. Furthermore, compared to matrices, tensors can more precisely capture the low-rank structure of stripe noise.

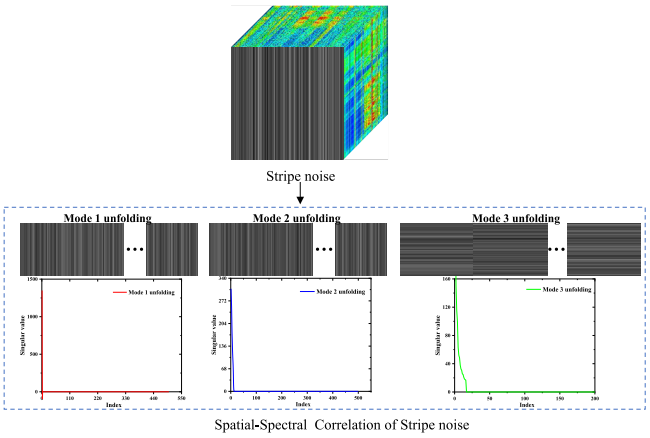


Fig. 1. Spatial low-rank priors of stripes.

Therefore, we propose to model the stripe noise $\mathcal{B}$ using tensors, extending the previous observation model into a double tensor model. The HSI observation model can then be rewritten as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} + \mathcal{B} \tag{2}$$

where $\mathcal{Y} \in \mathbb{R}^{h \times w \times p}$ denotes the observed HSI data; h and w denote the height and width of the ith band, respectively; and p denotes the number of bands.

C. Adaptive Hyper-Laplacian TV Prior

Based on the fact that image gradients present a sparse distribution, TV prior regularization and its extensions have been widely used in HSI denoising methods [19], among which SSTV takes into account both the spatial and spectral gradient information and demonstrates good performance. Specifically, SSTV can be written as

$$\begin{aligned}\|\mathcal{X}\|_{\text{s.t.}} = \sum_{i,j,k} & w_h |\mathcal{X}_{i,j,k} - \mathcal{X}_{i-1,j,k}| + w_w |\mathcal{X}_{i,j,k} - \mathcal{X}_{i,j-1,k}| \\ & + w_p |\mathcal{X}_{i,j,k} - \mathcal{X}_{i,j,k-1}|\end{aligned} \tag{3}$$

where (w_h, w_w, w_b) represent the weight along the three modes of $\mathcal{X}$ that controls its gradient sparsity, which are usually set to (1, 1, 0.5) [28]. SSTV regularization is implemented through the ℓ_1 norm, which results in the data distribution tending toward a Laplacian distribution, whose pdf is $p(x; \lambda) \propto e^{\lambda|x|}$. However, recent research has shown that the statistical distributions of both the spatial and spectral gradients are more consistent with a super-Laplacian distribution, with pdf $p(x; p) \propto ke^{-k|x|^p}$.

Fig. 2 shows the histograms of the gradient in three directions (spatial and spectral) and the fitting of the three different probability density curves: Gaussian, Laplacian, and super-Laplacian. As shown in Fig. 2, the super-Laplacian distribution can more effectively fit the distribution characteristics of the three gradient directions of the HSI. The super-Laplacian PDFs fit by different HSIs and different dimensions of the same HSI's gradient distribution characteristics will be different. Clearly, the closer the super-Laplacian pdf fit is to the real gradient distribution characteristic, the better the denoising effect.

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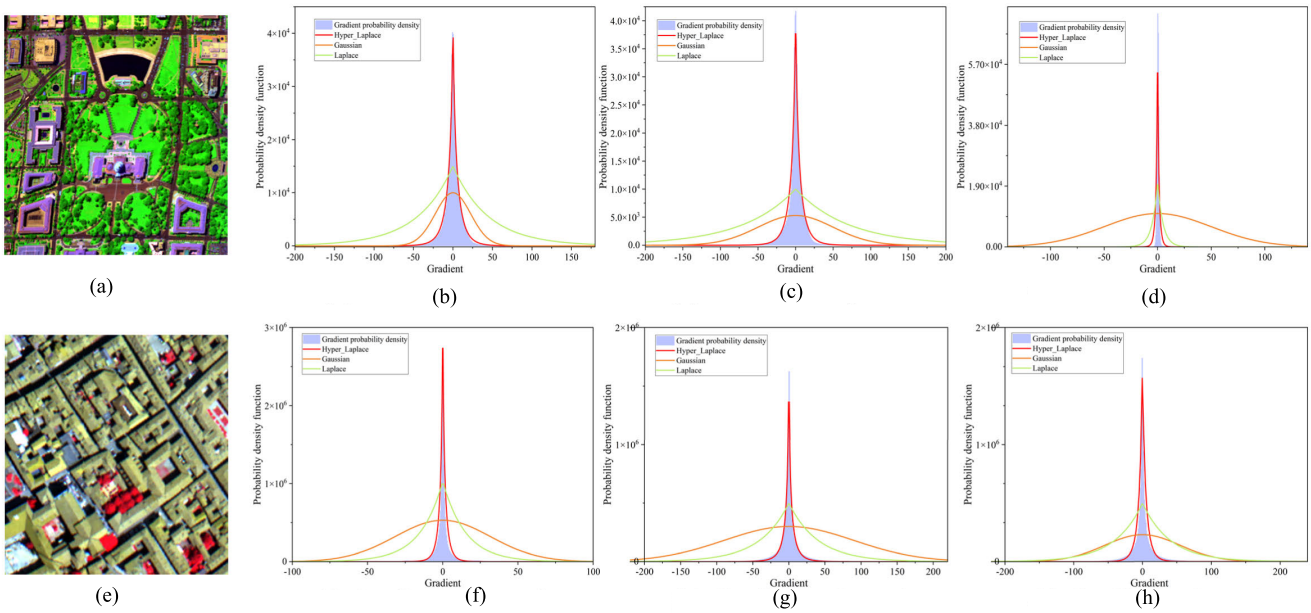


Fig. 2. Spatial–spectral gradient distributions and pdfs of the HSIs. (a) and (e) False-color images for the HSIs. (b)–(d) and (f)–(h) HSI histograms of gradients along the spatial and spectral dimensions and the fit probability density curves, respectively.

This means that, when designing a super-Laplacian regularizer, it is necessary to adaptively estimate the super-Laplacian parameters based on the gradient characteristics of the different dimensions of different images. By maximizing the likelihood function of the super-Laplacian distribution, it is possible to derive the AHSSTV term as follows:

$$\|\mathcal{X}\|_{\text{AHSSTV}} = w_h\|\mathcal{D}_x\mathcal{X}\|_{p_x}^{p_x} + w_w\|\mathcal{D}_y\mathcal{X}\|_{p_y}^{p_y} + w_b\|\mathcal{D}_z\mathcal{X}\|_{p_z}^{p_z} \tag{4}$$

where $\mathcal{D}_x$, $\mathcal{D}_y$, and $\mathcal{D}_z$ are the 3-D difference operators. p_x, p_y, and p_z are the 3-D hyper-Laplacian parameters. $p \in (0, 1)$, and when $p = 1$, AHSSTV degenerates to SSTV. We propose a method for adaptively estimating p based on the relevant research [43], [44].

1) For the observed HSI $\mathcal{Y}$, the clean image $\mathcal{X}$, and the noise $\mathcal{N}$, by performing differential operations in the spatial and spectral dimensions, three difference variables are constructed, $Y = [Y_i|i = 1, 2, 3] \in \mathbb{R}^{n_b\times p\times 3} Y_i = D_i(\mathcal{Y})$, $X = [X_i|i = 1, 2, 3] \in \mathbb{R}^{n_b\times p\times 3}$, $X_i = \mathcal{D}_i(\mathcal{X})$, $N = [N_i|i = 1, 2, 3] \in \mathbb{R}^{n_b\times p\times 3}$, $N_i = \mathcal{D}_i(\mathcal{N})$, where $\mathcal{D}_i(\cdot)$ is the difference operator in the ith dimension and n_b is the number of pixels in each spectral band. Based on the relevant research on noise estimation, a noise estimator is applied to Y to obtain the variance of the noise $\sigma = [\sigma_1, \sigma_2, \sigma_3]$, where $\sigma_i = \text{esn}(Y_i)$ and esn$(\cdot)$ is the noise variance estimator [46].
2) Since the clean image $\mathcal{Y}$ and noise $\mathcal{N}$ are independent of each other, it can be assumed that Y_i and N_i are mutually independent. At the same time, it can be assumed that Y_i and N_i follow a Laplacian distribution $f_{X_i} = k_i \exp(-k_i|X_i|^{p_i})$ and Gaussian distribution $f_{N_i} = (1/2\sqrt{\pi}\sigma_i)\exp(-N_i/4\sigma_i^2)$, respectively, and their pdf f_{Y_i} can be written as:

$$f_{Y_i} = f_{X_i} \times f_{N_i}. \tag{5}$$

The pdf of Y_i is

$$f_{Y_i}(Y_i = a) = \int_t f_{X_i}(X_i = t)\times f_{N_i}(N_i = a - t)dt. \tag{6}$$

Furthermore, due to the numerical discreteness of the image gradient, the probability density f with a normalized discrete histogram h can be approximated. The above equation can then be transformed into

$$h_{Y_i} = h_{X_i} \otimes h_{N_i} \tag{7}$$

where $\otimes$ represents the convolution operator, and h_{Y_i}, h_{N_i} can be calculated based on the observed HSI and noise variance. Furthermore, since X_i follows a Laplacian distribution, it is possible to obtain $h_{X_i} = h_f(k_i, p_i)$.

3) The goal is to estimate the optimal Laplacian distribution parameters $P = [p_i|i = 1, 2, 3]$ for the spatial–spectral gradient image in the 3-D, which can be written as

$$\{k_i, p_i\} = \arg\min_{k_i, p_i} \|h_{Y_i} - h_f(k_i, p_i) \otimes h_{N_i}\|_F^2 \tag{8}$$

where k_i and p_i are the parameters of the Laplacian distribution, but we only need parameter p_i in the derived Laplacian regularizer. Given that the gradient data of the image have a narrow range for both k_i and p_i, the Nelder-Mead algorithm [44] is used to search for these parameters and determine the optimal values. The impact of adaptive parameter p_i estimation on the denoising performance is further analyzed in Section III.

D. Optimization Model

As stated above, utilizing prior knowledge of the image to design an effective regularizer is the key to denoising

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HSIs. Based on the discussions in Sections III-A and III-C, we propose the LRTDAHL model

$$\begin{aligned}&\arg\min_{\mathcal{X},\mathcal{B},\mathcal{S}} \frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{B}\|_F^2+\lambda_1\|\mathcal{X}\|_{\text{AHSSTV}}+\lambda_2\|\mathcal{S}\|_1\\&\quad+\|\mathcal{X}\|_*+\|\mathcal{B}\|_*\\&\text{s.t.}\quad \mathcal{X}=\mathcal{C}\times_1 V_1\times_2 V_2\times_3 V_3, V_{\text{i}}^{\text{T}}V_{\text{i}}=I(i=1,2,3)\\&\quad\mathcal{B}=\mathcal{U}\times_1 G_1\times_2 G_2\times_3 G_3, G_{\text{i}}^{\text{T}}G_{\text{i}}=I(i=1,2,3)\end{aligned}\tag{9}$$

where λ_1 and λ_2 are the regularization parameters that control the adaptive hyper-Laplacian term and the ℓ_1 regularization term, respectively; $\|\cdot\|_*$ represents the low-rank tensor regularization term; $\|\cdot\|_{\text{AHSSTV}}$ represents the adaptive hyper-Laplacian regularization term; and $\|\cdot\|_1$ represents the ℓ_1 regularization term, which is used to remove the sparse noise. It is worth noting that the LRTDAHL model considers the prior information of the clean image, Gaussian noise, sparse noise, and stripe noise, and is expected to have the ability to remove multiple types of mixed noise. Specifically, we consider the prior information of the spatial–spectral low rank of the clean image and the spatial low rank of stripe noise, and introduce Tucker tensor low-rank decomposition to model them, which can fully encode the spatial–spectral low-rank characteristics of the clean HSI and the spatial low-rank characteristic of stripe noise. In addition, the AHSSTV regularization term can adaptively estimate the parameters required for the hyper-Laplacian regularization, which is better at preserving detailed information and suppressing Gaussian noise and sparse noise, compared to ℓ_1.

E. Optimization Procedure

As the proposed model is clearly a nonconvex optimization problem, there are various methods for solving nonconvex optimization problems. Based on the augmented Lagrange multiplier (ALM) optimization method (9), some auxiliary variables are first introduced to reformulate (9) as

$$\begin{aligned}&\arg\min_{\mathcal{X},\mathcal{B},\mathcal{S}} \frac{1}{2}\|\mathcal{Y}-\mathcal{Z}-\mathcal{S}-\mathcal{B}\|_F^2+\lambda_1\|\mathcal{F}\|_{\text{AHSSTV}}+\lambda_2\|\mathcal{S}\|_1\\&\quad+\|\mathcal{X}\|_*+\|\mathcal{B}\|_*+<\Lambda_1,\mathcal{X}-\mathcal{Z}>+<\Lambda_1,\mathcal{D}_w(\mathcal{Z})-\mathcal{F}\\&\quad>\frac{\beta}{2}\left(\|\mathcal{X}-\mathcal{Z}\|_F^2+\|\mathcal{D}_w(\mathcal{Z})-\mathcal{F}\|_F^2\right)\\&\text{s.t.}\quad \mathcal{Z}=\mathcal{X},\mathcal{D}_w(\mathcal{Z})=\mathcal{F},\mathcal{X}=\mathcal{C}\times_1 V_1\times_2 V_2\times_3 V_3\\&\quad V_{\text{i}}^{\text{T}}V_{\text{i}}=I(i=1,2,3),\mathcal{B}=\mathcal{U}\times_1 G_1\times_2 G_2\times_3 G_3\\&\quad G_{\text{i}}^{\text{T}}G_{\text{i}}=I(i=1,2,3)\end{aligned}\tag{10}$$

where Λ_1 and Λ_2 are the Lagrange multipliers, β is a positive penalty parameter, and $\mathcal{D}_w(\cdot)=[w_h\mathcal{D}_h(\cdot);w_w\mathcal{D}_w(\cdot);w_z\mathcal{D}_z(\cdot);]$ is a weighted difference operator. One solution to this problem is to iteratively optimize the augmented Lagrangian function on one variable while fixing the others, which is known as the alternating direction method of multipliers (ADMM) algorithm. Specifically, in the $k+1$th iteration, the variables are updated as follows.

1) Update $\mathcal{X}$:

$$\arg\min_{\mathcal{X}}\|\mathcal{X}\|_*+\frac{\beta}{2}\left\|\mathcal{X}-\frac{\beta\mathcal{Z}^k-\Lambda_1^k}{\beta}\right\|_F^2.\tag{11}$$

Tucker tensor decomposition is introduced, and (11) can be transformed into

$$\arg\min_{\mathcal{X}}\|\mathcal{X}\|_*+\frac{\beta}{2}\left\|\mathcal{C}\times_1 V_1\times_2 V_2\times_3 V_3-\frac{\beta Z^k-\Lambda_1^k}{\beta}\right\|_F^2.\tag{12}$$

By using the higher order orthogonal iteration (HOOI) algorithm [28], $\mathcal{C}^{k+1}$ and V_i^{k+1} can be obtained, and $\mathcal{X}$ can be updated as follows:

$$\mathcal{X}^{k+1}=\mathcal{C}^{k+1}\times_1 V_1^{k+1}\times_2 V_2^{k+1}\times_3 V_3^{k+1}.\tag{13}$$

2) Update $\mathcal{Z}$:

$$\begin{aligned}&\arg\min_{\mathcal{Z}}\frac{1}{2}\|\mathcal{Y}-\mathcal{Z}-\mathcal{S}^k-\mathcal{B}^k\|_F^2\\&\quad+\frac{\beta}{2}\left(\|\mathcal{X}^k-\mathcal{Z}\|_F^2+\|\mathcal{D}_w(\mathcal{Z})-\mathcal{F}^k\|_F^2\right)\\&\quad+<\Lambda_1^k,\mathcal{X}^k-\mathcal{Z}>+<\Lambda_2^k,\mathcal{D}_w(\mathcal{Z})-\mathcal{F}^k>.\end{aligned}\tag{14}$$

This problem can be viewed as a least-squares problem, and through derivation, the following can be obtained:

$$\begin{aligned}&\arg\min_{\mathcal{Z}}\frac{\beta+1}{2}\left\|\mathcal{Z}-\frac{\mathcal{Y}-\mathcal{B}^k-\mathcal{S}^k+\Lambda_1^{k+1}+\beta\mathcal{X}^{k+1}}{\beta+1}\right\|_F^2\\&\quad+\frac{\beta}{2}\left\|\mathcal{D}_w(\mathcal{Z})-\frac{\beta F^k-\Lambda_2^k}{\beta}\right\|_F^2.\end{aligned}\tag{15}$$

The items in (15) are expanded and similar terms are merged to obtain

$$\begin{aligned}\left(I+\beta+\beta\mathcal{D}_w^T\mathcal{D}_w\right)\mathcal{Z}&=\left(\mathcal{Y}-\mathcal{B}^k-\mathcal{S}^k+\Lambda_1^{k+1}+\beta\mathcal{X}^{k+1}\right)\\&\quad+\beta D_w^T\left(\mathcal{F}^k-\Lambda_2^k/\beta\right)\end{aligned}\tag{16}$$

where $\mathcal{D}_w^T$ is the companion matrix of $\mathcal{D}_w$, and $\mathcal{D}_w^T\mathcal{D}_w$ has a block matrix structure. To speed up the calculation, 3-D fast Fourier transform (FFT) can be used to solve the problem as follows:

$$\begin{cases}\text{H}_{\mathcal{Z}}=\left(\mathcal{Y}-\mathcal{B}^k-\mathcal{S}^k+\Lambda_1^{k+1}+\beta\mathcal{X}^{k+1}\right)\\\qquad+\beta D_w^T\left(\mathcal{F}^k-\Lambda_2^k/\beta\right)\\\text{T}_{\mathcal{Z}}=|\text{fftn}(\mathcal{D}_{\mathcal{X}})|^2+|\text{fftn}(\mathcal{D}_{\mathcal{Y}})|^2+|\text{fftn}(\mathcal{D}_{\mathcal{Z}})|^2\\Z^{k+1}=\text{ifftn}\left(\dfrac{\text{fftn}(\text{H}_{\mathcal{Z}})}{\beta+1+\beta\text{T}_{\mathcal{Z}}}\right)\end{cases}\tag{17}$$

where fftn$(\cdot)$ and ifftn$(\cdot)$ denote FFT and inverse FFT, respectively.

3) Update $\mathcal{F}$:

$$\arg\min_{\mathcal{F}}\lambda_1\|\mathcal{F}\|_{\text{AHSSTV}}+\frac{\beta}{2}\left\|\mathcal{F}-\frac{\beta\mathcal{D}_w\left(\mathcal{Z}^{k+1}\right)+\Lambda_2^k}{\beta}\right\|_F^2.\tag{18}$$

The following spatial–spectral difference operator is introduced:

$$L_\theta(\gamma,p_x,p_y,p_z)=\left[\|\gamma_{dx}\|_{p_x}^{p_x};\|\gamma_{dy}\|_{p_y}^{p_y};\|\gamma_{dz}\|_{p_z}^{p_z}\right]\tag{19}$$

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where γ_{dx}, γ_{dy}, and γ_{dz} represent the decoding operators of the 3-D of the spatial–spectral difference operator. $\|\cdot\|_p^p (0 < p < 1)$ is a nonconvex norm minimization problem, and the generalized iterative shrinkage algorithm [42] can lead to an accurate and convergent solution.

Therefore, the following can be obtained:

$$\mathcal{F}^{k+1} = L_{\frac{\lambda_1}{\beta}}\left(\frac{\beta \mathcal{D}_w\left(\mathcal{Z}^{k+1}\right) + \Lambda_2^k}{\beta}, p_h, p_w, p_p\right) \tag{20}$$

where p_h, p_w and p_p are the 3-D gradient hyper-Laplacian parameters estimated through (6) for the HSI.

4) Update $\mathcal{B}$:

$$\arg\min_{\mathcal{B}} \frac{\beta}{2}\|\mathcal{Y} - \mathcal{Z}^{k+1} - \mathcal{S}^k - \mathcal{B}\|_F^2 + \|\mathcal{B}\|_*. \tag{21}$$

By introducing Tucker tensor decomposition, (21) can be transformed into

$$\arg\min_{\mathcal{U},\mathrm{G}_i} \frac{\beta}{2}\|\mathcal{U} \times_1 G_1 \times_2 G_2 \times_3 G_3 - \left(\mathcal{Y} - \mathcal{Z}^{k+1} - \mathcal{S}^k\right)\|_F^2 + \|\mathcal{B}\|_*. \tag{22}$$

By using the HOOI algorithm, $\mathcal{U}^{k+1}$ and G_i^{k+1} can be obtained, and $\mathcal{B}$ can be updated as follows:

$$\mathcal{B}^{k+1} = \mathcal{U}^{k+1} \times_1 G_1^{k+1} \times_2 G_2^{k+1} \times_3 G_3^{k+1}. \tag{23}$$

5) Update $\mathcal{S}$:

$$\arg\min_{\mathcal{X},\mathcal{B},\mathcal{S}} \frac{1}{2}\|\mathcal{Y} - \mathcal{Z}^{k+1} - \mathcal{B}^{k+1} - \mathcal{S}\|_F^2 + \lambda_2\|\mathcal{S}\|_1. \tag{24}$$

The following soft threshold operator is then introduced:

$$\Re_\Delta(\mathcal{X}) = \begin{cases} \mathcal{X} - \Delta, & \mathcal{X} > \Delta \\ \mathcal{X} + \Delta, & \mathcal{X} < \Delta \\ 0, & \mathcal{X} = \Delta. \end{cases} \tag{25}$$

The closed-form solution of the problem (24) can be obtained by (25), i.e.,

$$S^{k+1} = \Re_\Delta\left(\mathcal{Y} - \mathcal{Z}^{k+1} - \mathcal{B}^{k+1}\right). \tag{26}$$

6) Update the Lagrangian:

$$\begin{cases} \Lambda_1^{k+1} = \Lambda_1^k + \beta\left(\mathcal{X}^{k+1} - \mathcal{Z}^{k+1}\right) \\ \Lambda_2^{k+1} = \Lambda_2^k + \beta\left(D_w\left(\mathcal{Z}^{k+1}\right) - \mathcal{F}^{k+1}\right). \end{cases} \tag{27}$$

Combining the subproblems solved above, the solution to the optimization process can be summarized as shown in Algorithm 1.

Algorithm 1 Optimization Procedure for the Proposed Model

1: **Input:** Noisy HSI $\mathcal{Y}$,Tucker desired rank [r_1,r_2,r_3] and [r_a,r_b, r_c], parameters $\varepsilon, \beta, \lambda_1, \lambda_2, k_{\max}$
2: **Initialize:** $\mathcal{X} = \mathcal{B} = \mathcal{S} = \mathcal{F} = \mathcal{Z} = 0$, $\Lambda_1 = \Lambda_2 = 0$, $\mu_{\max} = 10^6$, $k = 0$
3: **Estimate p:** Estimate p_h, p_w, p_p via (6)
4: **While** stopping criterion is not satisfied **do**:
5: Update $\mathcal{X}, \mathcal{C}, V_i$ via (12) and (13)
6: Update $\mathcal{Z}$ via (17)
7: Update $\mathcal{F}$ via (20)
8: Update $\mathcal{B}$ via (22) and (23)
9: Update $\mathcal{S}$ via (26)
10: Update the Lagrangian multipliers Λ_1, Λ_2 via (27)
11: Check the convergence condition $\frac{\|\mathcal{X}^{k+1} - \mathcal{X}^k\|_F^2}{\|\mathcal{X}^k\|_F^2} \le \varepsilon$
12: **Output:** The restored HSI

III. Experimental Results and Discussion

In this section, we describe how we tested the effectiveness of the proposed LRTDAHL model based on simulated and real-data experiments, and compared it with eight other state-of-the-art HSI denoising methods: band-wise asymmetric Laplacian (AL) noise-based matrix factorization (BALMF) [52], LRTDTV [28], a fast and parameter-free HSI mixed noise removal method (FastHyMix) [46], factor group sparsity-regularized nonconvex low-rank approximation (FGSLR) [47], empirical TPTV (ETPTV) [20], nonlocal meets global (NGmeet) [49], non-i.i.d. mixture of Gaussians (NMoG) [50], DLR matrix decomposition [3], fast graph laplacian regularization (FGLR) [51], nonlocal spatial–spectral neural network (NSSNN) [53], and correlated TV (CTV) [17]. For all the comparison methods, the parameter settings were adjusted according to the suggestions in the literature. The parameter selection for the proposed LRTDAHL model is discussed in Section III-C.

A. Simulated Data Experiments

1) Experimental Settings: To evaluate the noise removal performance of the proposed LRTDAHL model, we conducted simulated experiments and compared the visual and quantitative performances. Three representative clean HSI datasets were used in the experiments to simulate noisy HSIs, namely, the Washington DC Mall dataset, reflection optical system imaging spectrometer (ROSIS) Pavia City Center Dataset, and the Indian Pines dataset. There can be several different types of noise in real HSIs, including Gaussian noise, impulse noise, deadlines, and stripe noise. To simulate these scenarios as closely as possible, we compared six different simulated noise situations. Before the noise simulation, the gray values of each band were normalized to [0, 1].

To simulate noise in the HSIs, we designed the following six cases.

a) Case1 (Gaussian noise + impulse noise): Gaussian noise, deadlines, and impulse noise were added as illustrated in Table I, to assess the performance of the proposed method and the comparison methods in the absence of stripes.

b) Case2 (Gaussian noise + impulse noise + deadlines + stripes): Gaussian noise, deadlines, 40%–50% stripes, and impulse noise were added as illustrated in Table I, to assess the performance of the proposed method and the comparison methods under the condition of relatively low-intensity stripes.

c) Case3 (Gaussian noise + impulse noise + deadlines + stripes): Gaussian noise, deadlines, 60%–70% stripes, and impulse noise were added as illustrated in Table I, to assess the performance of the proposed method and the comparison methods under the condition of relatively high-intensity stripes.

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TABLE I
SIMULATED NOISE CASES

Simulated case	Added noise		
	σ^2 of Gaussian noise	Percentages of impulse noise	Intensity of stripes
Case1	0–0.2	0–0.2	0
Case2	0.1	0.2	40–50%
Case3	0–0.2	0–0.2	60–70%
Case4	0–0.2	0–0.2	40% periodic
Case5	0–0.2	0–0.2	40% mixed
Case6	0–0.2	0–0.2	Wide vertical stripes

TABLE II
QUANTITATIVE ASSESSMENT RESULTS OF ALL THE METHODS ON THE PAVIA CITY CENTER DATASET

Case	Evaluation index	Noisy	BALMF	NMoG	NGmeet	ETPTV	CTV	FGLR	NSSNN	DLR	FGSLR	FastHyMix	LRTDAHL
Case1	MPSNR	14.72	37.39	38.92	39.21	39.35	39.10	37.85	35.60	36.91	**40.14**	39.33	39.04
	MSSIM	0.261	0.976	0.979	0.988	0.984	0.980	0.961	0.955	0.969	**0.994**	0.992	0.980
	MSAM	0.723	0.090	**0.071**	0.088	0.072	0.083	0.091	0.096	0.115	0.079	0.087	0.081
Case2	MPSNR	10.05	33.82	33.57	32.88	35.47	35.35	34.91	33.21	35.58	35.47	30.98	**36.17**
	MSSIM	0.088	0.944	0.931	0.922	0.951	0.951	0.941	0.925	0.956	0.957	0.915	**0.966**
	MSAM	0.853	0.132	0.139	0.177	0.112	0.120	0.116	0.112	0.141	0.131	0.199	**0.104**
Case3	MPSNR	11.78	30.32	31.41	30.28	31.49	27.41	31.67	27.59	31.73	31.37	28.37	**32.79**
	MSSIM	0.171	0.894	0.943	0.906	0.923	0.893	0.931	0.897	0.943	0.944	0.867	**0.961**
	MSAM	0.875	0.205	0.195	**0.118**	0.146	0.231	0.148	0.164	0.232	0.144	0.187	0.122
Case4	MPSNR	12.24	26.19	27.92	28.12	29.36	27.83	28.75	27.68	30.11	27.29	25.32	**31.92**
	MSSIM	0.166	0.894	0.869	0.866	0.893	0.886	0.895	0.864	0.902	0.861	0.849	**0.924**
	MSAM	0.782	0.324	0.383	0.441	0.319	0.243	0.284	0.171	0.201	0.399	0.378	**0.195**
Case5	MPSNR	12.95	30.25	33.31	31.46	33.85	30.22	33.01	30.18	32.74	32.43	29.58	**34.98**
	MSSIM	0.182	0.894	0.904	0.882	0.910	0.896	0.907	0.902	0.896	0.894	0.870	**0.912**
	MSAM	0.842	0.286	0.214	0.222	0.199	0.209	0.201	0.178	0.196	0.214	0.353	**0.184**
Case6	MPSNR	14.22	29.85	30.94	32.83	31.35	30.81	31.72	30.92	32.79	31.62	29.37	**33.61**
	MSSIM	0.242	0.889	0.908	0.915	0.925	0.911	0.916	0.899	0.911	**0.938**	0.920	0.929
	MSAM	0.712	0.201	0.174	0.155	0.162	0.167	0.199	0.204	0.176	0.149	0.153	**0.146**

TABLE III
QUANTITATIVE ASSESSMENT RESULTS OF ALL THE METHODS ON THE INDIAN PINES DATASET

Case	Evaluation index	Noisy	BALMF	NMoG	NGmeet	ETPTV	CTV	FGLR	NSSNN	DLR	FGSLR	FastHyMix	LRTDAHL
Case1	MPSNR	14.44	38.96	38.30	40.44	38.10	38.58	37.41	35.34	35.81	**42.37**	39.66	41.05
	MSSIM	0.276	0.983	0.987	0.983	0.987	0.980	0.961	0.945	0.951	0.986	**0.989**	0.982
	MSAM	0.742	0.105	0.081	0.085	0.086	0.089	0.140	0.156	0.144	**0.079**	0.085	0.087
Case2	MPSNR	11.72	32.36	33.22	33.59	34.27	33.07	34.01	32.67	34.91	34.10	33.22	**35.59**
	MSSIM	0.167	0.931	0.951	0.957	0.949	0.937	0.949	0.939	0.954	0.941	0.921	**0.981**
	MSAM	0.759	0.192	0.161	0.198	0.167	0.201	0.211	0.253	0.221	0.197	0.241	**0.133**
Case3	MPSNR	13.01	30.75	33.23	32.46	32.35	30.83	33.21	31.24	32.71	33.15	32.51	**35.14**
	MSSIM	0.206	0.914	0.923	0.911	0.944	0.904	0.941	0.894	0.934	0.935	0.913	**0.962**
	MSAM	0.768	0.238	0.155	0.187	0.161	0.289	0.223	0.276	0.249	0.184	0.255	**0.139**
Case4	MPSNR	13.47	30.59	30.11	31.45	29.91	29.77	32.14	29.14	33.14	31.65	29.66	**36.11**
	MSSIM	0.241	0.912	0.931	0.899	0.878	0.874	0.932	0.867	0.951	0.904	0.876	**0.964**
	MSAM	0.788	0.194	0.186	0.211	0.222	0.217	0.194	0.271	0.188	0.178	0.289	**0.124**
Case5	MPSNR	13.56	29.46	32.25	29.24	32.33	31.70	33.01	29.73	33.21	33.14	32.64	**34.71**
	MSSIM	0.229	0.890	0.922	0.891	0.938	0.916	0.959	0.872	0.954	0.955	0.932	**0.969**
	MSAM	0.786	0.192	0.176	0.187	0.188	0.221	0.184	0.254	0.191	0.198	0.297	**0.165**
Case6	MPSNR	14.44	30.38	31.48	30.45	30.41	30.08	32.31	29.90	32.11	32.14	31.22	**33.40**
	MSSIM	0.271	0.907	0.931	0.905	0.949	0.908	0.951	0.898	0.941	0.947	0.939	**0.966**
	MSAM	0.741	0.199	0.201	0.181	0.195	0.231	0.201	0.242	0.223	0.196	0.214	**0.170**

d) Case4 (Gaussian noise + impulse noise + deadlines + periodic stripes): Gaussian noise, deadlines, 40% periodic stripes, and impulse noise were added as illustrated in Table I, to simulate the performance of the proposed method and the comparison methods in the presence of periodic stripes.

e) Case5 (Gaussian noise + impulse noise + deadlines + mixed stripes): Gaussian noise, deadlines, 40% mixed stripes, and impulse noise were added as illustrated in Table I, to simulate the performance of the proposed method and the comparison methods in the presence of mixed stripes.

f) Case6 (Gaussian noise + impulse noise + deadlines + wide vertical stripes): Gaussian noise, deadpixels, wide vertical stripes, and impulse noise were added as illustrated in Table I, to simulate the performance of the proposed method

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TABLE IV
QUANTITATIVE ASSESSMENT RESULTS OF ALL THE METHODS ON THE WASHINGTON DC MALL DATASET

Case	Evaluation index	Noisy	BALMF	NMoG	NGmeet	ETPTV	CTV	FGLR	NSSNN	DLR	FGSLR	FastHyMix	LRTDAHL
Case1	MPSNR	14.12	36.97	37.88	40.11	37.66	37.64	37.59	35.81	35.57	**40.32**	39.96	39.83
	MSSIM	0.231	0.945	0.954	**0.974**	0.963	0.947	0.938	0.915	0.919	0.971	0.961	0.964
	MSAM	0.702	0.165	0.124	0.111	0.134	0.149	0.141	0.154	0.161	**0.098**	0.102	0.101
Case2	MPSNR	11.80	30.44	31.99	30.54	31.64	29.99	32.14	29.14	31.61	32.93	29.88	**34.41**
	MSSIM	0.109	0.897	0.901	0.891	0.919	0.896	0.922	0.876	0.191	0.914	0.892	**0.933**
	MSAM	0.787	0.191	0.184	0.198	0.161	0.207	0.151	0.230	0.164	0.156	0.198	**0.114**
Case3	MPSNR	13.95	29.32	30.17	28.77	28.45	27.05	31.04	28.57	30.91	29.84	29.13	**32.41**
	MSSIM	0.181	0.899	0.894	0.889	0.895	0.876	0.919	0.872	0.901	0.891	0.887	**0.946**
	MSAM	0.683	0.201	0.207	0.221	0.235	0.244	0.181	0.232	0.191	0.201	0.212	**0.169**
Case4	MPSNR	14.35	27.05	29.54	27.66	27.86	29.04	30.59	27.11	30.41	29.47	28.31	**32.18**
	MSSIM	0.208	0.878	0.899	0.866	0.874	0.891	0.911	0.873	0.902	0.901	0.876	**0.922**
	MSAM	0.730	0.267	0.231	0.246	0.253	0.231	0.247	0.262	0.231	0.234	0.219	**0.201**
Case5	MPSNR	13.32	30.63	32.02	29.55	32.98	30.86	31.48	28.66	31.74	32.11	30.29	**33.62**
	MSSIM	0.182	0.906	0.907	0.877	0.911	0.895	0.909	0.881	0.899	0.901	0.904	**0.916**
	MSAM	0.651	0.213	0.204	0.214	0.199	0.218	0.214	0.219	0.201	0.196	0.201	**0.194**
Case6	MPSNR	14.10	30.03	32.84	30.11	30.75	30.94	32.91	29.02	32.94	31.47	31.45	**34.12**
	MSSIM	0.229	0.895	0.911	0.895	0.913	0.906	0.917	0.894	0.907	0.914	0.916	**0.925**
	MSAM	0.678	0.195	0.116	0.157	0.152	0.169	0.134	0.209	0.140	0.122	0.124	**0.115**

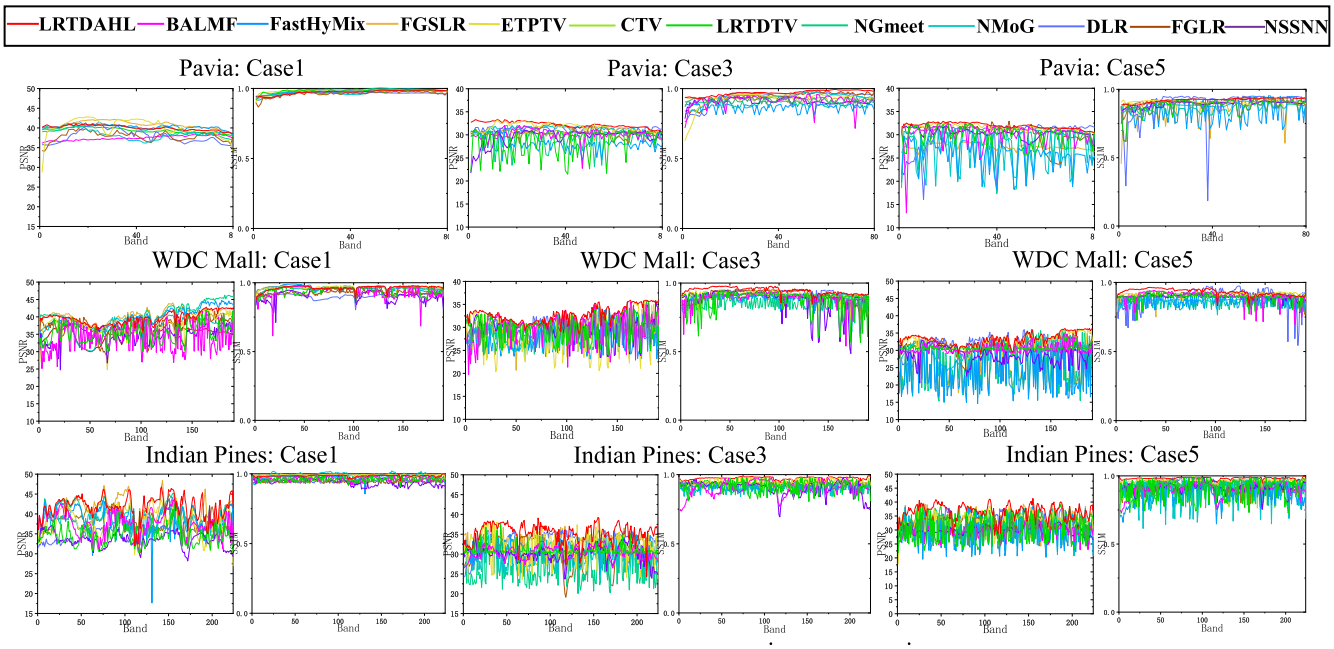


Fig. 3. PSNR and SSIM values of each band of the restored results obtained by the different methods.

and the comparison methods in the presence of wide vertical stripes.

2) Quantitative Comparison: The quantitative evaluation results of the various methods on the simulated experimental datasets under the different noise conditions are presented in Tables II–IV.

Tables II–IV present the mean peak signal-to-noise ratio (MPSNR), mean structural similarity index measure (MSSIM), and mean spectral angle mapper (MSAM) values obtained for all the denoising methods on the Pavia City Center, Indian Pines, and Washington DC Mall datasets. The best results for each case are highlighted in bold and the second-best results are underlined. It can be seen that the proposed LRTDAHL method exhibits a better overall performance than the other comparison methods in removing mixed noise, particularly in the presence of strong stripe noise or specific stripes. In Case1 (absence of stripe noise), the proposed method is very close to the optimal results. The noise modeling-based NMoG method and the nonconvex low-rank approximation-based FGSLR method exhibit slight advantages or comparability to the proposed method in cases of only Gaussian noise and sparse noise, but their performance is inferior to that of LRTDAHL in the presence of stripe noise. Overall, these quantitative comparisons demonstrate the effectiveness of the proposed method in HSI denoising and stripe removal.

Fig. 3 presents a comparison of the peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) values for various bands in Case1, Case3, and Case5 of the simulated experiments. The results indicate that the performance of NSSNN is not very robust. Although the proposed method does not show significant advantages over the state-of-the-art methods in Case1, it exhibits considerable advantages

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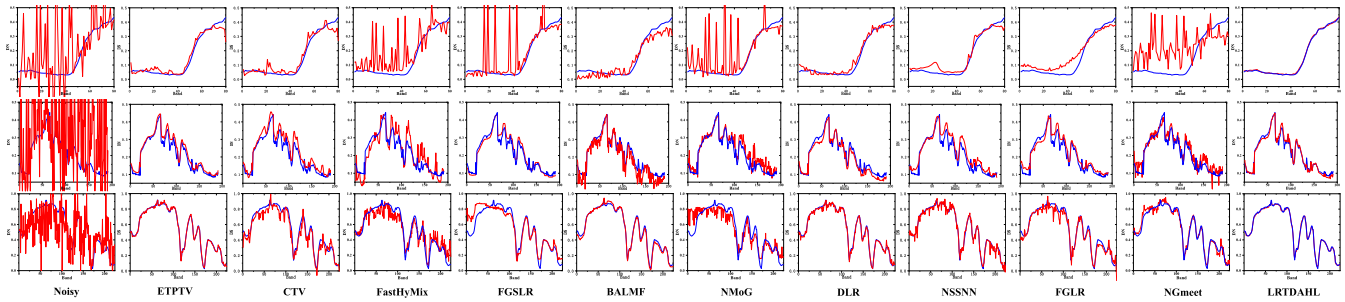


Fig. 4. Spectral curves of the restored results obtained by the different methods. The red curves are the denoising results and the blue curves are the original spectral curves. The first row shows the results obtained at spatial location (134, 76) of the Pavia City Center dataset under Case6. The second row shows the results obtained at spatial locations (107, 214) of the Washington DC Mall dataset under Case6. The last row shows the results obtained at spatial location (30, 30) of the Indian Pines dataset under Case6.

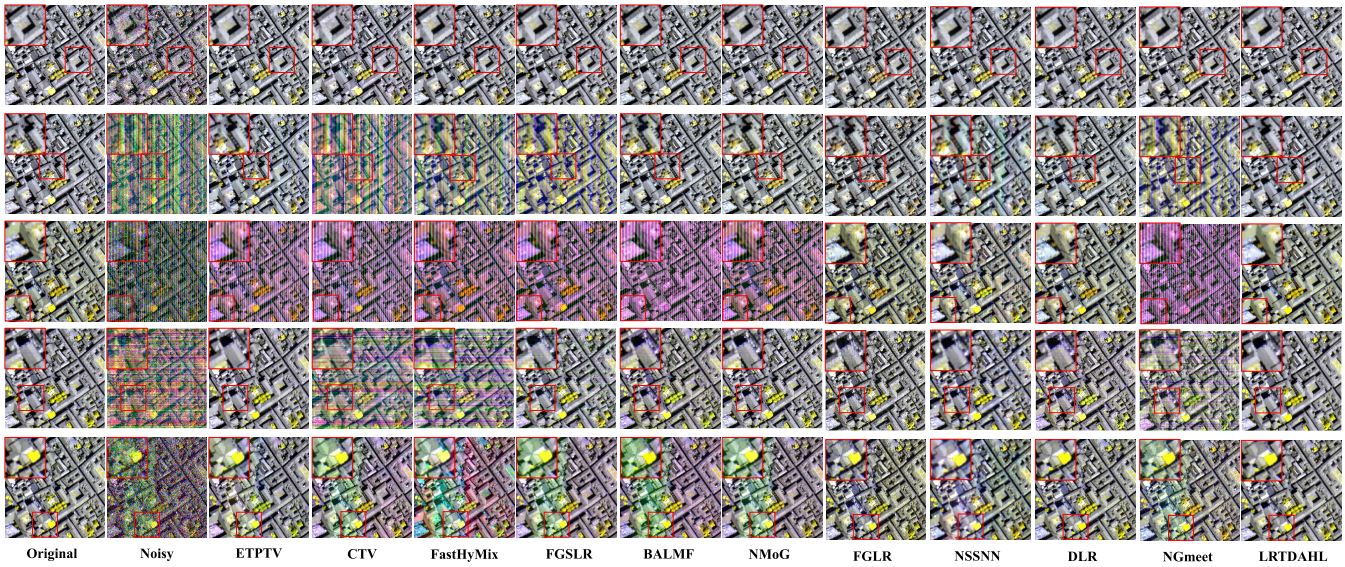


Fig. 5. Denoising false-color results obtained on the Pavia City Center dataset by the different methods. From the first row to the last row are Case1, Case3, Case4, Case5, and Case6, respectively.

in almost all the bands when high-intensity stripe noise and specific stripes are present.

To assess the spectral recovery capability of each method, Fig. 4 illustrates the clean spectra (blue curves) and denoised spectra (red curves) of pixels in the simulated datasets. It is evident that the mixed noise has disrupted the spectral characteristics of the pixels, and FastHyMix, NMOG, and NGMeet struggle to recover the spectral curves effectively. DLR, ETPTV, and BALMF can reconstruct the spectral curves well but still exhibit some significant fluctuations. In contrast, our proposed method achieves the closest recovery to the real spectral curves.

3) Qualitative Comparison: Further performance comparisons of the nine comparison methods were made, and their results were visually analyzed. Specifically, the Pavia City Center, Indian Pines, and Washington DC Mall datasets were used to evaluate the denoising performance of each method under the Case1, Case3, Case4, Case5, and Case6 scenarios, as shown in Figs. 5–7. The subregions marked in each subplot are enlarged and framed in red boxes for better visual clarity. From these figures, it can be observed that all methods have more or less removed the mixed noise. The proposed method, compared to others, achieves significant denoising performance while preserving image textures effectively.

Based on the results and analysis of the above-simulated experiments, it can be concluded that BALMF and NMoG, as state-of-the-art noise modeling methods, exhibit effective removal of randomly distributed mixed noise in Case3. However, their performance in removing regular stripe noise is relatively weak. NGMeet and FastHyMix, representing tensor subspace methods, demonstrate fast processing speeds (refer to Section III-C4 for details). NGMeet, being a state-of-the-art Gaussian denoiser, performs exceptionally well in the Gaussian noise case. However, it struggles to remove stripe noise in the presence of mixed noise. FastHyMix can eliminate a significant portion of mixed noise but still leaves residual stripes, especially in the case of wide stripes. NSSNN, as an advanced deep learning method, exhibits fast processing speeds and performs well in handling Gaussian noise. However, its performance across different cases is unstable, primarily due to its relatively weak generalization ability. ETPTV and CTV represent state-of-the-art TV-related methods. CTV exhibits fast processing speeds, but its ability to handle dense mixed noise is compromised due to limited consideration of noise priors. On the other hand, ETPTV can remove the majority of noise, but residual noise remains when dealing with periodic and wide stripes. DLR is an advanced method that considers modeling stripe noise and effectively addresses various types

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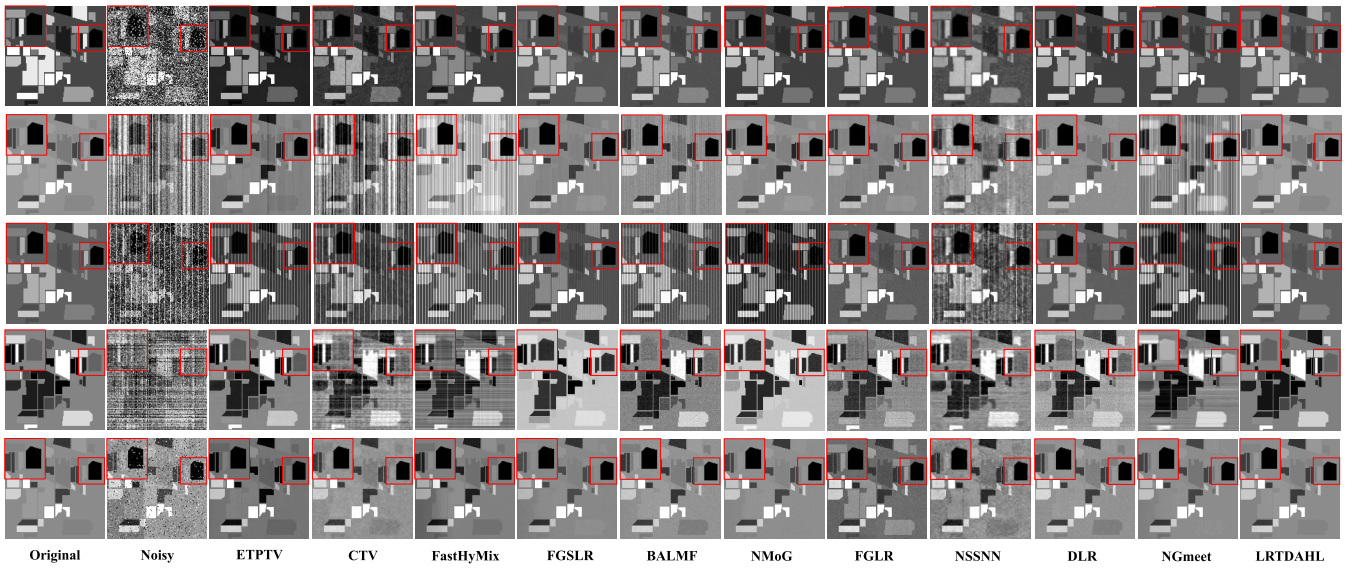


Fig. 6. Denoising results obtained on the Indian Pines dataset by the different methods. From the first row to the last row are Case1, Case3, Case4, Case5, and Case6, respectively.

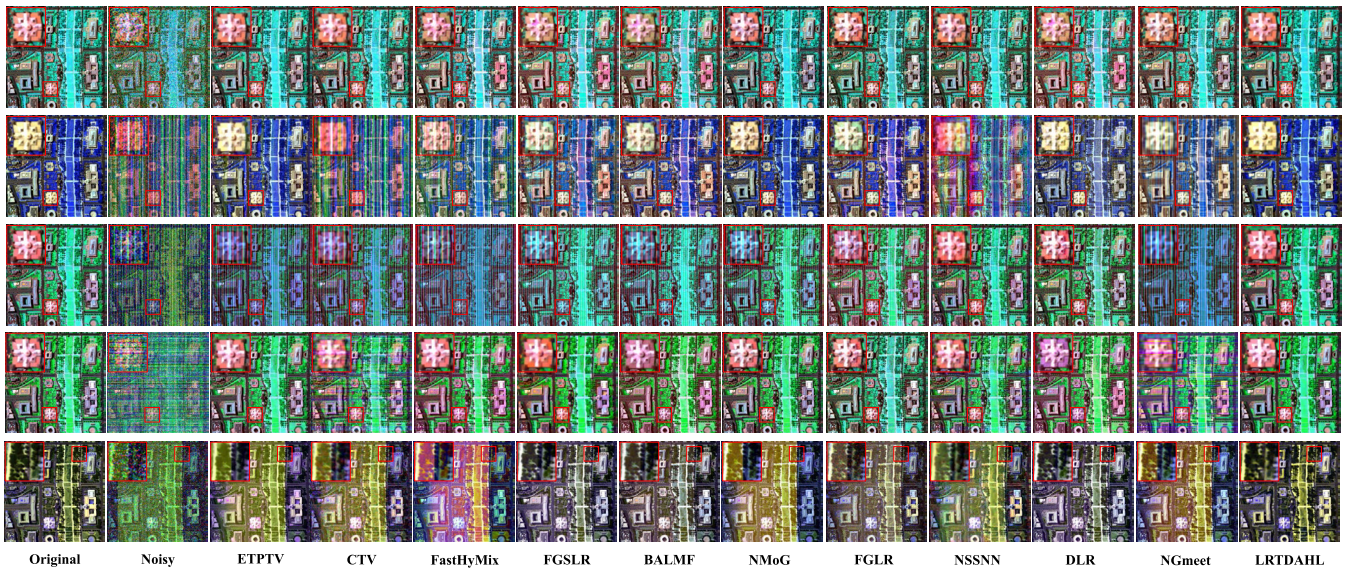


Fig. 7. Denoising false-color results obtained on the Washington DC Mall dataset by the different methods. From the first row to the last row are Case1, Case3, Case4, Case5, and Case6, respectively.

of stripe noise. However, there is still residual Gaussian noise when handling mixed noise. FGLR and FGSLR are representative low-rank methods. FGLR exhibits extremely fast processing speeds, but its performance is not very effective when dealing with mixed noise. FGSLR, on the other hand, almost achieves optimal performance in Case1 and performs well in handling mixed noise, yet residual periodic and wide stripes remain. Our proposed method demonstrates results very close to the optimal in Case1 and effectively addresses various types of stripe noise in Case2–6, almost achieving optimal performance.

B. Real-Data Experiments

The effectiveness of the proposed model was tested using three real datasets, namely, the Hyperspectral Digital Imagery Collection Experiment (HYDICE) Urban dataset, an Earth Observing-1 (EO-1) Hyperion dataset, and a Gaofen-5 (GF-5) dataset. Prior to applying the denoising model, the HSIs were normalized band-by-band to the range of [0, 1].

1) HYDICE Urban Dataset: The HYDICE Urban dataset is a publicly available and widely used dataset in hyperspectral remote sensing data recovery experiments. It was acquired by the HYDICE sensor and contains a total of 210 bands ranging from 0.4 to 2.4 μm in the visible and near-infrared (VNIR) spectral range. The dataset is severely contaminated by Gaussian noise, deadlines, stripe noise, water absorption noise, atmospheric effects, and some unknown noise.

Figs. 8 and 9 present the denoised images of the 108th and 150th bands of the HYDICE Urban dataset obtained using different methods. The images are shown before and after

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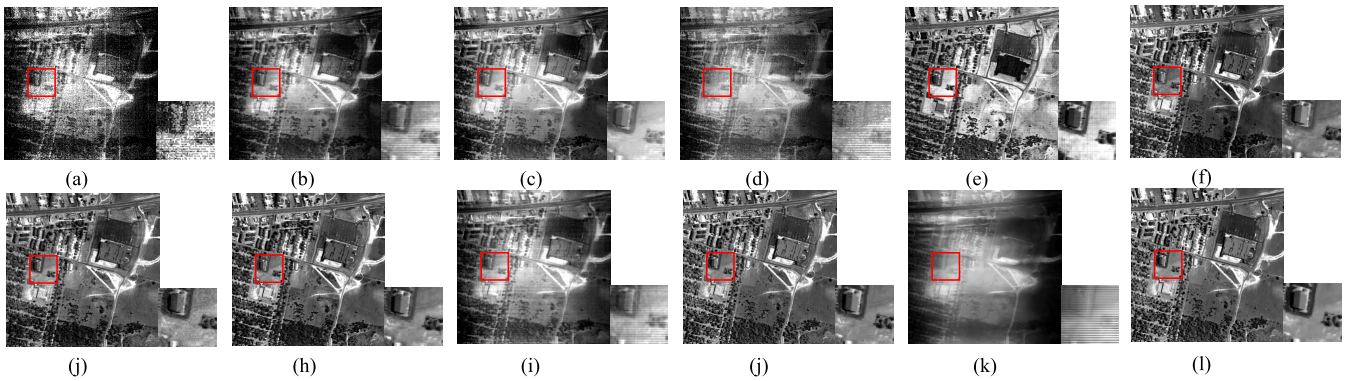


Fig. 8. Denoised results for band 108 of the HYDICE Urban dataset obtained by the different methods. (a) Original. (b) CTV. (c) ETPTV. (d) BALMF. (e) FastHyMix. (f) DLR. (g) FGLR. (h) FGSLR. (i) NSSNN. (j) NMoG. (k) NGmeet. (l) LRTDAHL.

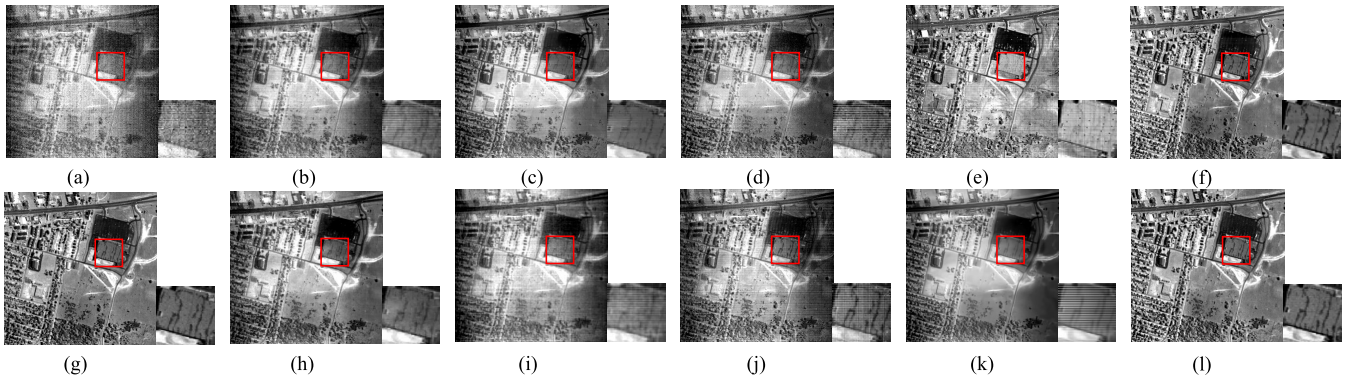


Fig. 9. Denoised results for band 139 of the HYDICE Urban dataset obtained by the different methods. (a) Original. (b) CTV. (c) ETPTV. (d) BALMF. (e) FastHyMix. (f) DLR. (g) FGLR. (h) FGSLR. (i) NSSNN. (j) NMoG. (k) NGmeet. (l) LRTDAHL.

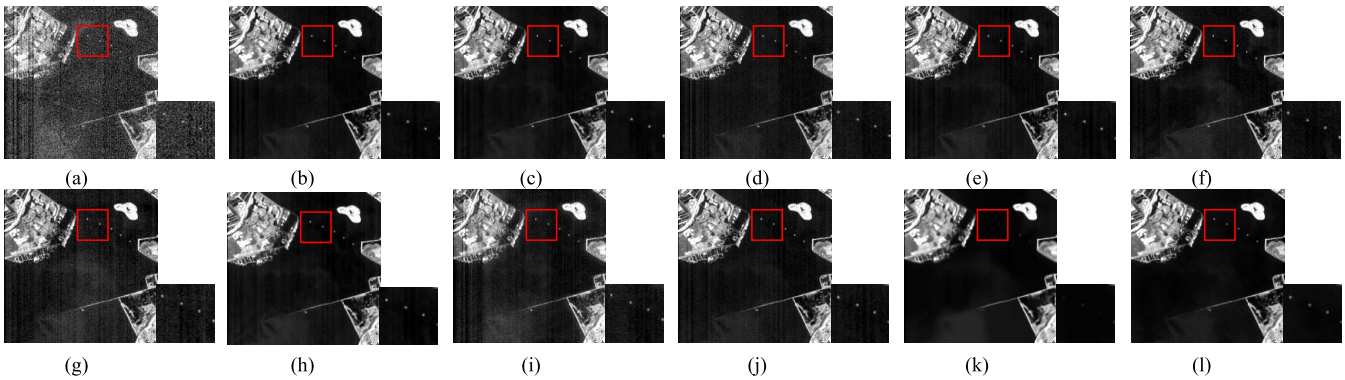


Fig. 10. Denoising results for band 130 of the EO-1 dataset obtained by the different methods. (a) Original. (b) CTV. (c) ETPTV. (d) BALMF. (e) FastHyMix. (f) DLR. (g) FGLR. (h) FGSLR. (i) NSSNN. (j) NMoG. (k) NGmeet. (l) LRTDAHL.

denoising, with local zoomed-in detail displayed in the red boxes below each image. The original images are severely corrupted by mixed noise, including dense stripes, Gaussian noise, and impulse noise. It can be observed that the NGMeet, CTV, and BALMF methods only remove a small amount of the mixed noise, and cannot eliminate the dense noise near the image edges and the wide stripe noise, which can cause significant image distortion. The ETPTV and FastHyMix methods exhibit excellent performance in removing the Gaussian and sparse noise but perform poorly in removing the stripe noise, and the zoomed-in detail reveals the presence of significant stripe noise even after denoising. Visually, FGSLR, NMoG, DLR, and the proposed LRTDAHL method demonstrate the best performances, effectively removing the mixed noise while preserving the spatial details. The narrow distribution of the noise-contaminated bands, the concentrated contamination intensity, and the lack of prominent striping noise contamination contribute to the relatively easier separation between noise and clean signals in this dataset, which could be a potential explanation for the comparable results obtained by the proposed method and the state-of-the-art approaches.

2) EO-1 Hyperion Dataset: This dataset was obtained by the Hyperion hyperspectral imaging spectrometer, spanning 400–2500 nm across a total of 242 bands, with a spectral resolution of 10 nm. However, the dataset is severely corrupted by various forms of noise, including Gaussian noise, dead

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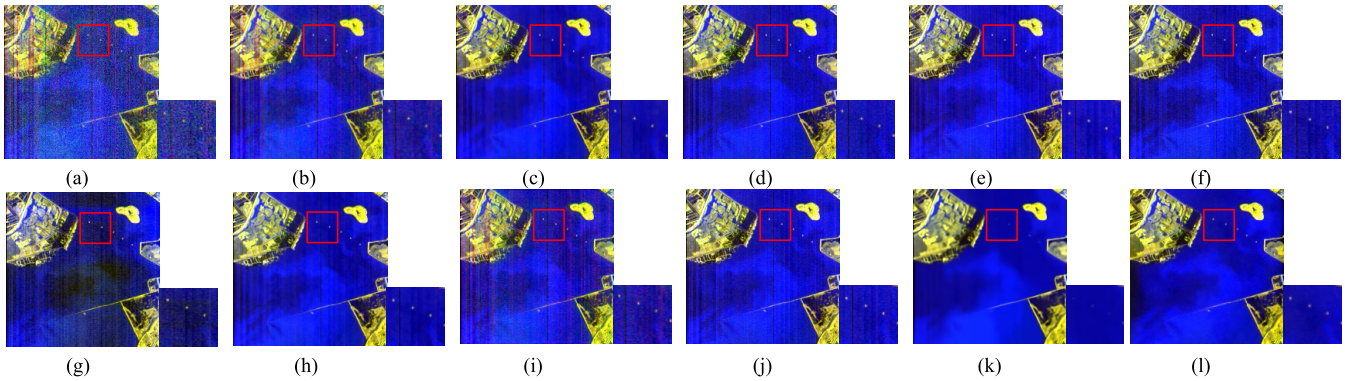


Fig. 11. Denoised false-color results for the EO-1 dataset obtained by the different methods. (a) Original. (b) CTV. (c) ETPTV. (d) BALMF. (e) FastHyMix. (f) DLR. (g) FGLR. (h) FGSLR. (i) NSSNN. (j) NMoG. (k) NGmeet. (l) LRTDAHL.

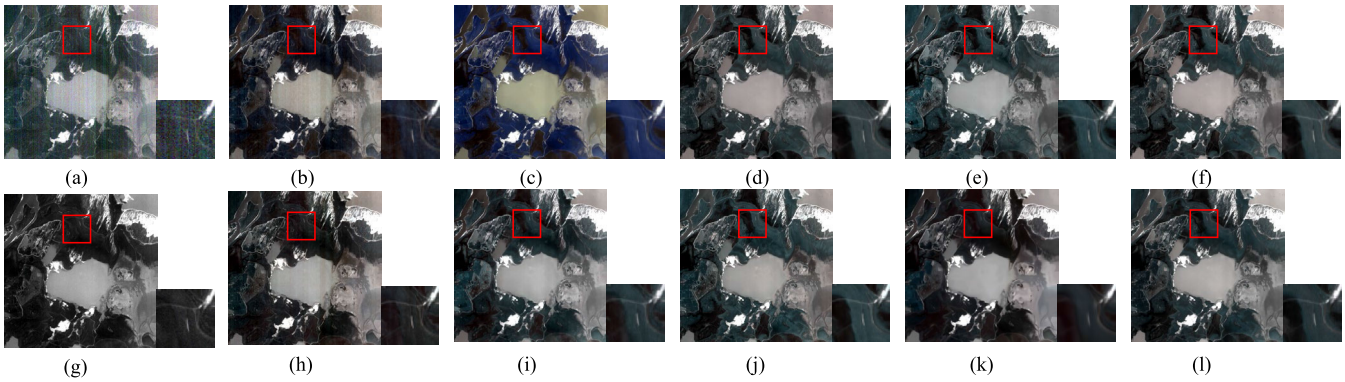


Fig. 12. Denoised false-color results for the GF5 VN dataset obtained by the different methods. (a) Original. (b) CTV. (c) ETPTV. (d) BALMF. (e) FastHyMix. (f) DLR. (g) FGLR. (h) FGSLR. (i) NSSNN. (j) NMoG. (k) NGmeet. (l) LRTDAHL.

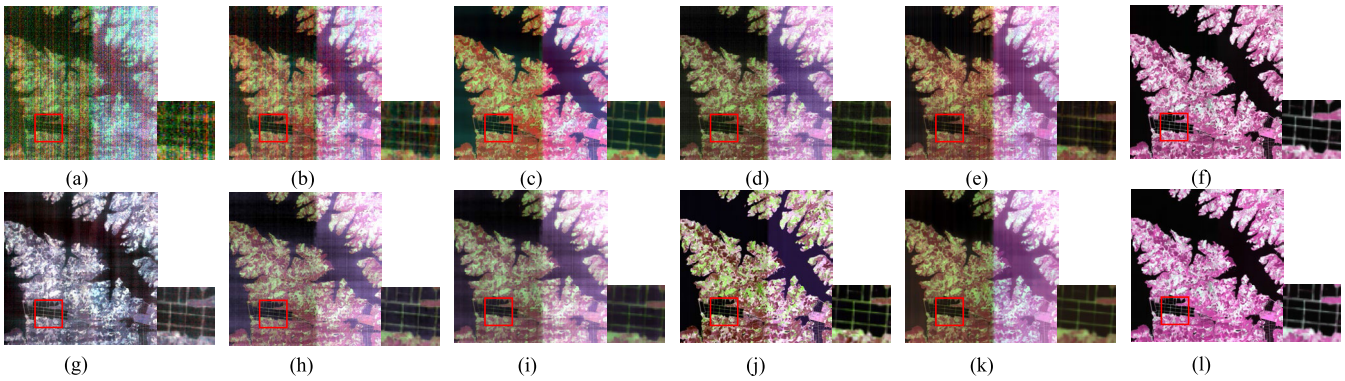


Fig. 13. Denoised false-color results for the GF5 dataset obtained by the different methods. (a) Original. (b) CTV. (c) ETPTV. (d) BALMF. (e) FastHyMix. (f) DLR. (g) FGLR. (h) FGSLR. (i) NSSNN. (j) NMoG. (k) NGmeet. (l) LRTDAHL.

pixels, stripe noise, water absorption noise, atmospheric noise, and other unidentified noise sources.

Figs. 10 and 11 present the restoration results for the EO-1 dataset obtained through the different methods. The original image is contaminated by stripe noise, Gaussian noise, sparse noise, and dead pixels. The BM4D method exhibits weak noise removal capabilities. Although NGmeet performs well in noise reduction, it generates excessive smoothing and loses local details. The results obtained through DLR and FGLR processing still exhibit subtle remnants of stripe artifacts. The other comparison methods can eliminate most of the Gaussian and sparse noise, but stripe noise and noticeable deadlines remain. Overall, the proposed LRTDAHL method demonstrates the best performance in terms of noise removal.

3) GF-5 Dataset: The GF-5 dataset was acquired using the advanced hyperspectral imager (AHSI) instrument, and consists of a total of 330 bands ranging from 400 to 2500 nm. The GF-5 dataset is divided into visible near infrared (VN) and short wave infrared (SW) datasets. The GF-5 SW dataset is contaminated by mixed noise, encompassing stripe noise, Gaussian noise, sparse noise, and dead pixels. Similarly, the GF-5 VN dataset is affected by mixed noise, but the contamination intensity is lower than that of the GF5 SW dataset and is concentrated in the initial few bands.

From Fig. 12, it can be observed that all methods are capable of effectively removing noise from the GF5 VN dataset. However, ETPTV tends to introduce noticeable spectral distortions, while NGMeet exhibits some degree of oversmoothing.

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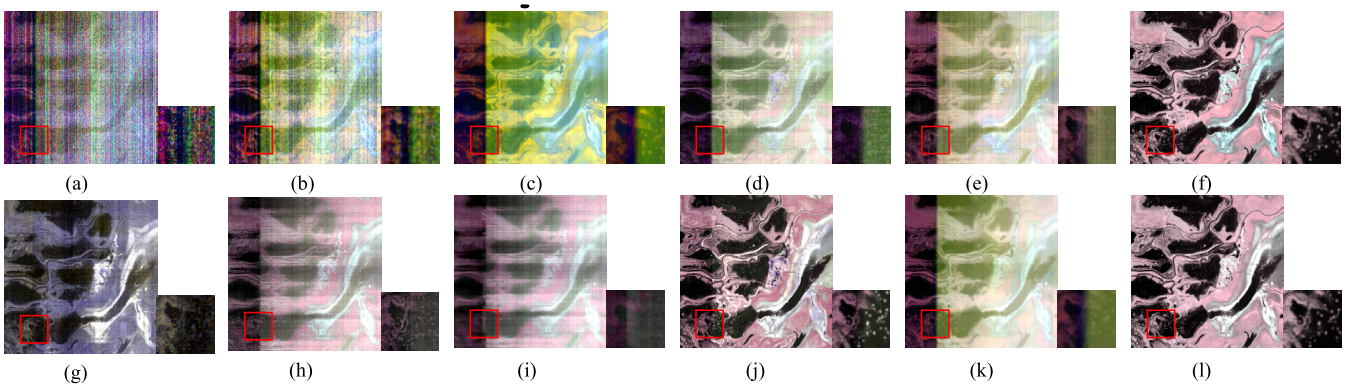


Fig. 14. Denoised false-color results for the GF5 dataset obtained by the different methods. (a) Original. (b) CTV. (c) ETPTV. (d) BALMF. (e) FastHyMix. (f) DLR. (g) FGLR. (h) FGSLR. (i) NSSNN. (j) NMoG. (k) NGmeet. (l) LRTDAHL.

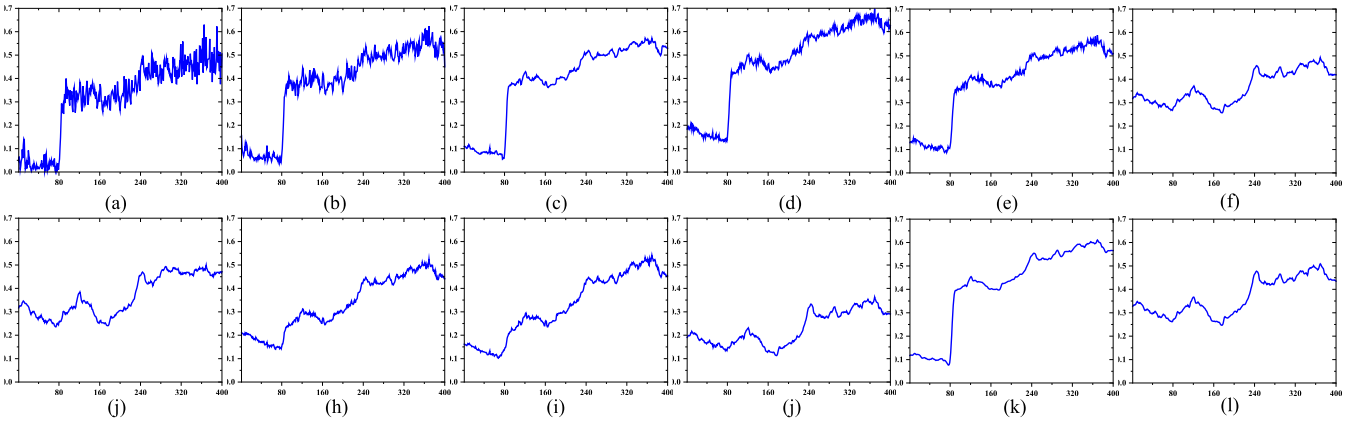


Fig. 15. Mean vertical profiles of band 52 in the GF5 SW dataset obtained by the different methods. (a) Original. (b) CTV. (c) ETPTV. (d) BALMF. (e) FastHyMix. (f) DLR. (g) FGLR. (h) FGSLR. (i) NSSNN. (j) NMoG. (k) NGmeet. (l) LRTDAHL.

From Figs. 13 and 14, it can be observed that the ETPTV, FastHyMix, and FGSLR methods can remove most of the noise in GF5 SW dataset but still leave significant stripe artifacts and produce over-smoothing. NMoG, FGLR, and NGmeet methods demonstrate better performance, but still leave stripe noise artifacts and cause spectral distortion. Overall, the proposed LRTDAHL method and the DLR method can effectively remove the mixed noise and demonstrate the best performance.

Fig. 15 illustrates the column mean curve before and after processing for the 52 band of the GF5 SW dataset. This particular band is heavily contaminated with mixed noise, including wide stripes. Specifically, the first 80 columns are densely polluted by low-value stripes, resulting in an overall lower mean value for these columns. It can be observed that all methods smooth the column mean curve to some extent. However, only DLR, NMoG, FGLR, and the proposed method are capable of correcting the underestimation distortion caused by wide stripes in the first 80 columns, but the column mean curve after FGLR processing still exhibits some degree of oscillation. Overall, the proposed method demonstrates the best smoothing effect on the column mean curve and effectively corrects anomalies.

C. Discussion

1) Parameter Analysis: The proposed LRTDAHL method's performance in denoising HSIs is affected by the selection of regularization parameters λ_1 and λ_2 and rank values r_1 and r_2. In this study, we investigated the influence of these parameters on the proposed method's performance.

The impact of parameter variations in the denoising of the simulated noisy images from Case1 to Case6 was investigated. Specifically, the effect of λ_1 and λ_2 on the denoising results was analyzed. λ_1 is the contribution parameter for the SSTV constraint, while λ_2 is the regularization parameter for sparse noise. The proposed method was tested with varying λ_1 and λ_2 values, which were selected from the ranges of [0, 0.001, 0.002, 0.003, 0.004, 0.005, 0.006, 0.007] and [0, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07], respectively.

The presented model's performance on the simulated Pavia City Center dataset with varying values of parameters λ_1 and λ_2 is illustrated in Fig. 15. In Case1, from Fig. 16(a), it can be seen that when the noise types are limited to Gaussian noise and sparse noise, the proposed method achieves optimal MPSNR values in the vicinity of λ_1 ranging from 0.001 to 0.003 and λ_2 ranging from 0 to 0.02. In Case2 to Case6, as illustrated in Fig. 16(b)–(f), when adding stripe noise, a not very low value of λ_1 can bring better results. The proposed approach achieves optimal MPSNR values in the vicinity of λ_1 ranging from 0.002 to 0.007 and λ_2 ranging from 0.02 to 0.07. Both excessively small and large values of λ_1 lead to diminished denoising results, with λ_2 exhibiting a relatively lower sensitivity, compared to λ_1. Based on this observation, λ_1 and λ_2 were set to 0.002 and 0.02, respectively, in both the simulated and real-data experiments conducted in this study.

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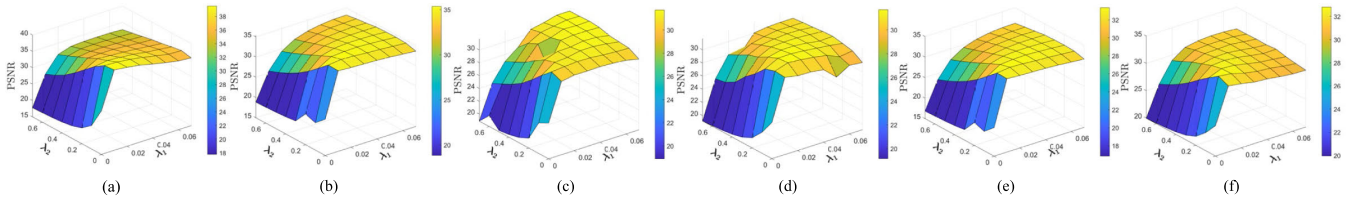


Fig. 16. Sensitivity analysis for parameters λ_1 and λ_2 in the simulated experiments. (a) Case1. (b) Case2. (c) Case3. (d) Case4. (e) Case5. (f) Case6.

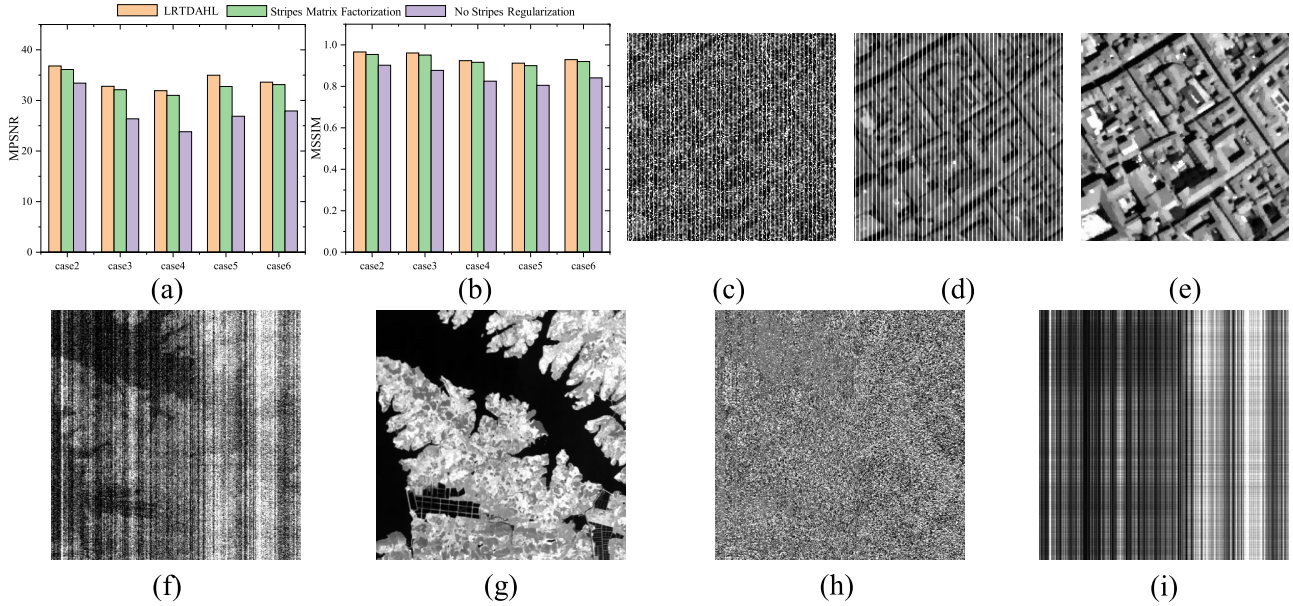


Fig. 17. Effectiveness of the stripe tensor decomposition. The results for Case2 to Case6 for LRTDAHL with stripe matrix decomposition and no stripe tensor decomposition. (a) MPSNR. (b) MSSIM. The results for band 40 in Case4. (c) Noisy. (d) No stripe tensor decomposition. (e) LRTDAHL result. The results for band 180 in the GF-5 dataset. (f) Original. (g) LRTDAHL result. (h) Gaussian and sparse noise. (i) Stripe noise.

2) Effectiveness of the Stripe Tensor Decomposition Term: In this section, we discuss the superiority of the stripe tensor decomposition term in the LRTDAHL model.

The quantitative results of the experiments on the simulated Pavia City Center dataset are shown in Fig. 17(a) and (b), where the original images were contaminated with noise similar to Case2 to Case6. We compared the performance of LRTDAHL with stripe tensor decomposition and stripe matrix decomposition separately, and it can be observed that the performance is worse when the stripe noise is not modeled separately. Using tensor decomposition to characterize stripes performs better than using matrix decomposition, as tensor decomposition can simultaneously encode the high correlation of the stripe components in the two spatial directions and the correlation in part of the spectral direction, as opposed to the per-band matricization. Fig. 17(c)–(e) show the image results with and without stripe tensor decomposition, and it can be observed that, without applying the low-rank tensor term for stripes, it is impossible to completely separate the periodic stripes in the spatial direction from the image components. Fig. 17(f)–(i) show the decomposition components of the GF-5 dataset in band 180 after denoising with LRTDAHL, and it can be visually observed that the proposed method can separate the clean imagery, sparse noise, Gaussian noise, and stripe noise, which potentially demonstrates the effectiveness of the two tensor decomposition terms in LRTDAHL.

3) Effectiveness of the Adaptive Hyper-Laplacian Term: In this section, the effectiveness of the adaptive hyper-Laplacian term in the LRTDAHL model is discussed.

As is well known, different HSIs often exhibit varying gradient distributions, which results in the density functions of the optimal hyper-Laplacian distributions they conform to being dissimilar. In this context, Fig. 18 presents the experimental results obtained on the simulated Pavia City Center dataset, where the original images were contaminated with the same noise as in Case2. Specifically, Fig. 18(a)–(c) depict the quantitative results of the denoising performance after varying p_h, p_w, and p_p while keeping the other parameters constant. The proposed LRTDAHL method obtains a good result with its estimated p values, with the results consistently near the optimal values. In addition, Fig. 18(e)–(g) present the denoising results for the false-color images obtained with different p_h values while keeping the other parameters constant. The value $p_h = 0.642$ is the result of the adaptive estimation. It can be observed that excessively large p values lead to residual noise, while excessively small p values lead to overly smooth results with lost image details. In contrast, the proposed method's estimated p values exhibit a good performance.

4) Convergence of the LRTDAHL Solver: It is evident that the proposed model (9) is a nonconvex optimization problem, and nonconvex optimization problems are difficult to find globally optimal solutions for and prove their convergence.

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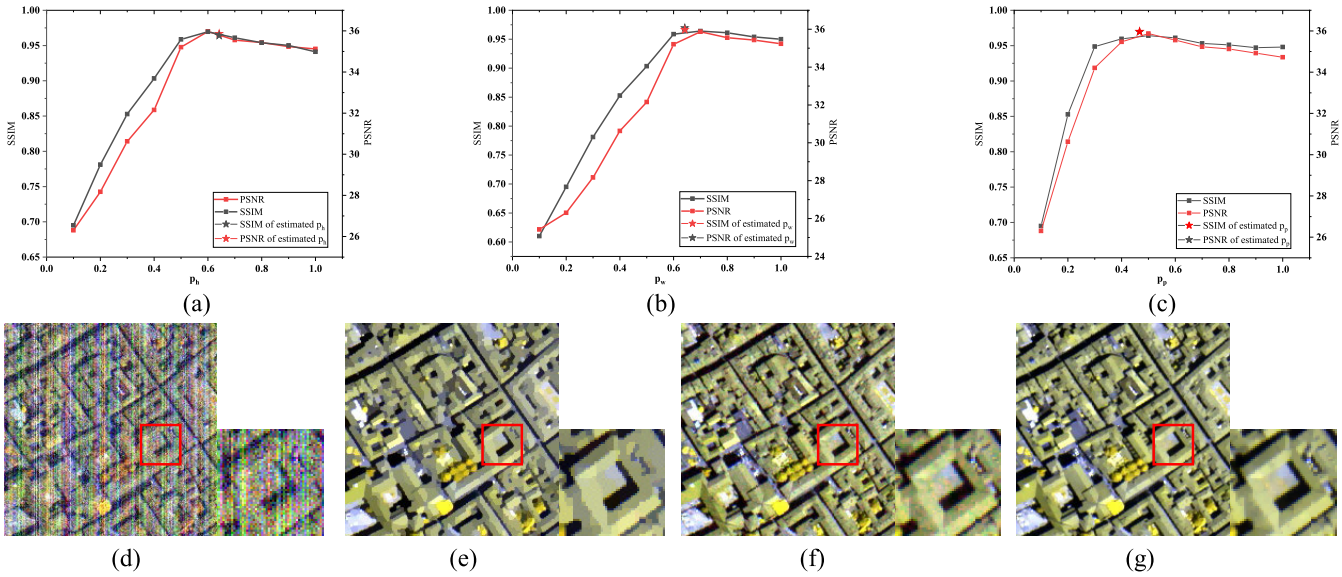


Fig. 18. Effectiveness analysis for the adaptive hyper-Laplacian prior. (a) Performance with different p_h values for the hyper-Laplacian prior (estimated $p_h = 0.642$). (b) Performance with different pw values for the hyper-Laplacian prior (estimated $p_w = 0.684$). (c) Performance with different pp values for the hyper-Laplacian prior (estimated $p_p = 0.485$). (d) Noisy false-color image of Case5. (e) False-color image for $p_h = 0.3$. (f) False-color image for $p_h = 1.0$. (g) False-color image for $p_h = 0.642$.

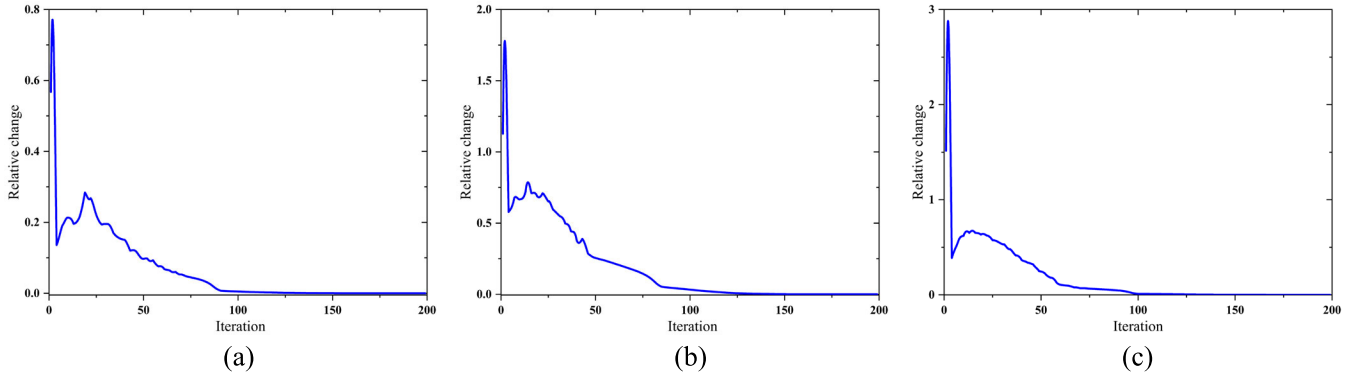


Fig. 19. Relative change values of LRTDAHL with respect to the iteration number. (a) HYDICE Urban dataset. (b) EO-1 dataset. (c) GF-5 dataset.

TABLE V
RUNNING TIME COMPARISON (IN SECONDS)

Case	BALMF	NMoG	NGmeet	ETPTV	CTV	FGLR	NSSNN	DLR	FGSLR	FastHyMix	LRTDAHL
WDC	263.7	119.7	66.9	128.2	71.4	6.4	4.3	64.6	1404.3	3.2	549.3
GF5	708.4	233.3	143.9	318.4	159.2	7.7	6.4	71.7	1567.1	3.9	913.9

Therefore, we introduce the ALM method to optimize the proposed model. Fig. 19 shows the evolution curve of the relative change with iteration number in the real-data experiments. It can be observed that the relative change converges to zero when the iteration number reaches a high value, indicating that the convergence of LRTDAHL can be guaranteed.

5) Computational Speeds: As shown in Table V, we compared the runtime performance of different models on both a simulated dataset and a real-world dataset. It can be observed from the results that FastHyMix and NSSNN are the fastest among all the compared methods. However, as indicated in the previous experiments, it is evident that they do not effectively eliminate complex types of stripe noise. The FGSLR method incurs a longer runtime in comparison to other methods, primarily due to the execution of a computationally expensive iterative algorithm at each iteration. For the proposed LRTDAHL algorithm in this article, its runtime is relatively higher compared to other methods for removing mixed noise, such as CTV, DLR, and BALMF. The increased runtime is attributed to the necessity of performing the HOOI algorithm twice during the iteration process, involving multiple SVD operations. Improving the algorithm's runtime speed is a focus for our future work.

IV. CONCLUSION

In this article, we have proposed the tensor-based LRTDAHL method for HSI denoising and destriping. Specifically, we utilize tensor low-rank decomposition to characterize

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the spatial–spectral correlation of the HSI and stripe noise and introduce adaptive hyper-Laplacian regularization to further encode the spatial–spectral information of the HSI gradients. Extensive simulation and real-data experiments were conducted, and the results clearly demonstrated that the proposed LRTDAHL method outperformed the other classical and state-of-the-art methods in both the visual and quantitative evaluations.

In the future, we plan to explore the utilization of the forward pass of a three-layer implicit neural network [54] as a substitute for the HOOI algorithm. This approach aims to mitigate the computational burden associated with SVD operations, consequently accelerating the processing speed of the model.

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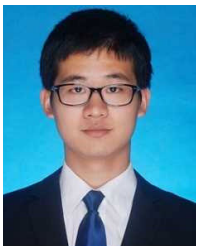
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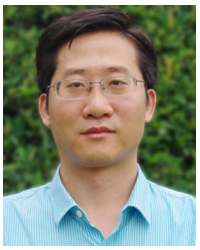
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